

Exercise Solutions: For Loop

1. Display First 10 Natural Numbers

Solution

```
public class FirstTenNaturalNumbers {  
    public static void main(String[] args) {  
        System.out.println("The first 10 natural numbers are:");  
  
        for (int number = 1; number <= 10; number++) {  
            System.out.println(number);  
        }  
    }  
}
```

Sample Output

```
The first 10 natural numbers are:  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

2. Sum and Average of 5 Numbers

Solution

```
import java.util.Scanner;

public class SumAndAverage {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int sum = 0;

        System.out.println("Input the 5 numbers :");
        for (int count = 1; count <= 5; count++) {
            int number = input.nextInt();
            sum += number;
        }

        double average = sum / 5.0;
        System.out.println("The sum of 5 no is : " + sum);
        System.out.println("The Average is : " + average);

        input.close();
    }
}
```

Sample Output

```
Input the 5 numbers :
1
2
3
4
5
The sum of 5 no is : 15
The Average is : 3.0
```

3. Display Cubes of Numbers

Solution

```
import java.util.Scanner;

public class DisplayCubes {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number of terms : ");
        int terms = input.nextInt();

        for (int number = 1; number <= terms; number++) {
            int cube = number * number * number;
            System.out.println("Number is : " + number
                               + " and cube of " + number + " is : " + cube);
        }

        input.close();
    }
}
```

Sample Output

```
Input number of terms : 4
Number is : 1 and cube of 1 is : 1
Number is : 2 and cube of 2 is : 8
Number is : 3 and cube of 3 is : 27
Number is : 4 and cube of 4 is : 64
```

4. Multiplication Table of a Number

Solution

```
import java.util.Scanner;

public class MultiplicationTable {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input the number (Table to be calculated) : ");
        int number = input.nextInt();
        System.out.print("Input number of terms : ");
        int terms = input.nextInt();

        for (int multiplier = 0; multiplier <= terms; multiplier++) {
            int result = number * multiplier;
            System.out.println(number + " X " + multiplier + " = " + result);
        }

        input.close();
    }
}
```

Sample Output

```
Input the number (Table to be calculated) : 5
Input number of terms : 5
5 X 0 = 0
5 X 1 = 5
5 X 2 = 10
5 X 3 = 15
5 X 4 = 20
5 X 5 = 25
```

5. Sum of Odd Natural Numbers

Solution

```
import java.util.Scanner;

public class SumOfOddNumbers {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Input number of terms is: ");
        int terms = input.nextInt();
        int sum = 0;

        System.out.println("The odd numbers are :");
        for (int count = 1; count <= terms; count++) {
            int oddNumber = 2 * count - 1;
            System.out.println(oddNumber);
            sum += oddNumber;
        }

        System.out.println("The Sum of odd Natural Number upto "
            + terms + " terms is: " + sum);

        input.close();
    }
}
```

Sample Output

```
Input number of terms is: 5
The odd numbers are :
1
3
5
7
9
The Sum of odd Natural Number upto 5 terms is: 25
```